Chapter 9, Problem 19P

Problem

Using phasors, find:

(a) 3 cos(20*t* + 10°) − 5 cos(20*t* − 30°)

(b) 40 sin 50*t* + 30 cos(50*t* − 45°)

(c) 20 sin 400*t* + 10 cos(400*t* + 60°) −5 sin(400*t* − 20°)

Step-by-step solution

Step 1 of 3

Refer to the section 9.3 in the textbook.

(a)

Consider the expression, $v(t)$.

$$v(t) = 3\cos(20t + 10°) - 5\cos(20t - 30°)$$

Convert the expression into phasor form.

$$\begin{aligned} V &= 3\angle 10° - 5\angle -30° \\ &= 3(\cos 10° + j\sin 10°) - 5(\cos(-30°) + j\sin(-30°)) \\ &= 3(0.9848 + j0.1736) - 5(0.866 + j(-0.5)) \\ &= (2.954 + j0.521) - (4.33 - j2.5) \end{aligned}$$

$$V = -1.376 + j3.021$$

Convert the rectangular form to polar form.

$$\begin{aligned} V &= \sqrt{(-1.376)^2 + (3.021)^2}\angle \tan^{-1}\left(-\frac{3.021}{1.376}\right) \\ &= 3.32\angle -65.51°\text{ V} \end{aligned}$$

Or

$$V = 3.32\angle 114.49° \qquad (\text{since, } 180° - 65° = 114.49°)$$

Step 2 of 3

Write the phasor in time domain.

$$v(t) = 3.32\cos(20t - 65.51°)$$

Or

$$v(t) = 3.32\cos(20t + 114.49°)$$

Hence, the required resultant expression is,

$$\boxed{3.32\cos(20t - 65.51°)\ (\text{Or})\ 3.32\cos(20t + 114.49°)}.$$

Step 3 of 3

(b)

Consider the expression, $v(t)$.

$$\begin{aligned} v(t) &= 40\sin 50t + 30\cos(50t - 45°) \\ &= 40\cos(50t - 90°) + 30\cos(50t - 45°) \end{aligned}$$

Convert the expression into phasor form.

$$\begin{aligned} \text{V} &= 40\angle -90° + 30\angle -45° \\ &= 40\left(\cos(-90°) + j\sin(-90°)\right) + 30\left(\cos(-45°) + j\sin(-45°)\right) \\ &= 40\left(0 + j(-1)\right) + 30\left(0.707 + j(-0.707)\right) \\ &= -j40 + 21.213 - j21.213 \end{aligned}$$

$$= 21.213 - j61.213$$

Convert the rectangular form to polar form.

$$\begin{aligned} V &= \sqrt{(21.213)^2 + (-61.213)^2}\angle \tan^{-1}\left(-\frac{61.213}{21.213}\right) \\ &= 64.784\angle -70.886° \end{aligned}$$

Write the phasor in time domain.

$$v(t) = 64.78\cos(50t - 70.886°)$$

Hence, the required resultant expression is, $\boxed{64.78\cos(50t - 70.886°)}$.